

**Multiple Choice:**

1. Approximately how much gain does a half-wave dipole have over an isotropic radiator?
 - A. 1.5 dB
 - B. 6.0 dB
 - C. 3.0 dB
 - D. 2.1 dB
2. What is a parasitic beam antenna?
 - A. An antenna where the driven element obtains its radio energy by induction or radiation from director elements
 - B. An antenna where all elements are driven by direct connection to the feed line
 - C. An antenna where some elements obtain their radio energy by induction or radiation from a driven element
 - D. An antenna where wave traps are used to magnetically couple the elements
3. If the forward gain of a six- element Yagi is about 10 dB, what would the gain of two of these antennas be if they were "stacked"?
 - A. 7 dB
 - B. 20 dB
 - C. 13 dB
 - D. 10 dB
4. The spacing between the elements on a three-element Yagi antenna, representing the best overall choice, is _____ of a wavelength.
 - A. 0.15
 - B. 0.75
 - C. 0.5
 - D. 0.2
5. A dipole transmitting antenna, placed so that the ends are pointing North/South, radiates:
 - A. mostly to the South and North
 - B. mostly to the South
 - C. equally in all directions
 - D. mostly to the East and West
6. The cubical "quad" or "quad" antenna consists of two or more square loops of wire. The driven element has an approximate overall length of:
 - A. three-quarters of a wavelength
 - B. one wavelength
 - C. two wavelengths
 - D. one-half wavelength
7. What is one effect of increasing the boom length and adding directors to a Yagi antenna?
 - A. SWR increases
 - B. Weight decreases
 - C. Wind load decreases
 - D. Gain increases
8. What does "antenna front-to- back ratio" mean in reference to a Yagi antenna?
 - A. The relative position of the driven element with respect to the reflectors and directors
 - B. The power radiated in the major radiation lobe compared to the power radiated in exactly the opposite direction
 - C. The power radiated in the major radiation lobe compared to the power radiated 90 degrees away from that direction
 - D. The number of directors versus the number of reflectors
9. What is one effect of increasing the boom length and adding directors to a Yagi antenna?
 - A. SWR increases
 - B. Weight decreases
 - C. Wind load decreases
 - D. Gain increases
10. Which statement about two-element delta loops and quad antennas is true?
 - A. They perform very well only at HF
 - B. They compare favorably with a three element Yagi
 - C. They are effective only when constructed using insulated wire
 - D. They perform poorly above HF
11. What is meant by antenna gain?
 - A. The numerical ratio of the signal in the forward direction to the signal in the back direction
 - B. The numerical ratio of the amount of power radiated by an antenna compared to the transmitter output power
 - C. The final amplifier gain minus the transmission line losses
 - D. The numerical ratio relating the radiated signal strength of an antenna to that of another antenna
12. How many directly driven elements do most Yagi antennas have?
 - A. None
 - B. Three
 - C. Two
 - D. One
13. How does the bandwidth of a folded dipole antenna compare with that of a simple dipole antenna?
 - A. It is essentially the same
 - B. It is less than 50%
 - C. It is 0.707 times the bandwidth
 - D. It is greater
14. What is a good way to get maximum performance from a Yagi antenna?
 - A. Optimize the lengths and spacing of the elements
 - B. Use RG-58 feed line
 - C. Use a reactance bridge to measure the antenna performance from each direction around the antenna
 - D. Avoid using towers higher than 9 meters (30 feet) above the ground
15. In free space, what is the radiation characteristic of a half-wave dipole?
 - A. Minimum radiation from the ends, maximum broadside
 - B. Maximum radiation from the ends, minimum broadside
 - C. Omnidirectional
 - D. Maximum radiation at 45 degrees to the plane of the antenna
16. How can the bandwidth of a parasitic beam antenna be increased?
 - A. Use traps on the elements



- B. Use larger diameter elements
C. Use tapered-diameter elements
D. Use closer element spacing
17. If a magnetic-base whip antenna is placed on the roof of a car, in what direction does it send out radio energy?
- A. Most of it is aimed high into the sky
B. Most of it goes equally in two opposite directions
C. It goes out equally well in all horizontal directions
D. Most of it goes in one direction
18. Why is a 5/8-wavelength vertical antenna better than a 1/4-wavelength vertical antenna for VHF or UHF mobile operations?
- A. A 5/8-wavelength antenna has less corona loss
B. A 5/8-wavelength antenna has more gain
C. A 5/8-wavelength antenna is easier to install on a car
D. A 5/8-wavelength antenna can handle more power
19. What happens to the feed point impedance of a ground-plane antenna when its radials are changed from horizontal to downward-sloping?
- A. It increases
B. It decreases
C. It stays the same
D. It approaches zero
20. In free space, what is the radiation characteristic of a half-wave dipole?
- A. Minimum radiation from the ends, maximum broadside
B. Maximum radiation from the ends, minimum broadside
C. Omnidirectional
D. Maximum radiation at 45 degrees to the plane of the antenna
21. Which of the following transmission lines will give the best match to the base of a quarter-wave ground-plane antenna?
- A. 300 ohms balanced feed line
B. 75 ohms balanced feed line
C. 300 ohms coaxial cable
D. 50 ohms coaxial cable
22. The front-to-back ratio of a beam antenna is:
- A. the forward power of the major lobe to the power in the backward direction both being measured at the 3 dB points
B. the ratio of the maximum forward power in the major lobe to the maximum backward power radiation
C. undefined
D. the ratio of the forward power at the 3 dB points to the power radiated in the backward direction
23. The property of an antenna, which defines the range of frequencies to which it will respond, is called its:
- A. Bandwidth
B. front-to-back ratio
C. impedance
D. polarization
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- A. 300 ohms balanced feed line
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END